

Instructor: Date:

Learner: Regional Branch:

Program: Center:

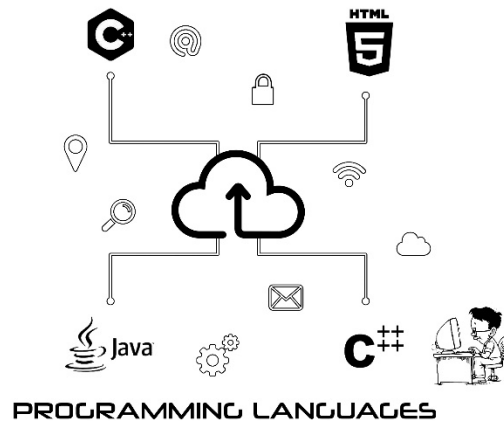
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Let's Start!

Imperative Languages in Software Programming



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can talk about the imperative languages in software programming.
- I can list some imperative languages.
- I can give examples of imperative languages.

Ask your partner:

- What are your favorite programming languages? Why?
- Do you think there is a common command programming languages use? Why?



Let's Explore!

Mark is hosting a radio show in which a special guest is explaining what imperative programming is.



Instructions: Read the text below.

Mark: For today's program, we decided to bring the imperative languages paradigm in computer science into discussion. So, Kim, a software programmer, agreed to join us and help us to understand this topic. Hi, Kim, tell us. What are imperative languages in software programming?

Kim: Thank you, Mark. Well, imperative languages are programming languages that work under the imperative paradigm, that is, the way we design programs. The imperative paradigm is very important because it enables all elemental computer functionalities to work.

Mark: Interesting! And, what is the imperative paradigm about?

Kim: The imperative paradigm focuses on the processes the program performs rather than on the goals that the program needs to achieve. Imperative languages describe how the program has to work; for that reason, an imperative programming language is, basically, a list of commands for the computer to perform. Consequently, Imperative programming consists of statements that modify the state of a program. That's why programmers fix logic statements orderly and descriptively for the computer to execute.

Mark: So, I suppose it's really taxing to program all those statements.

Kim: It used to be tiring, but with the advent of procedural programming, all became faster.

Mark: Is procedural programming different from the imperative programming paradigm?

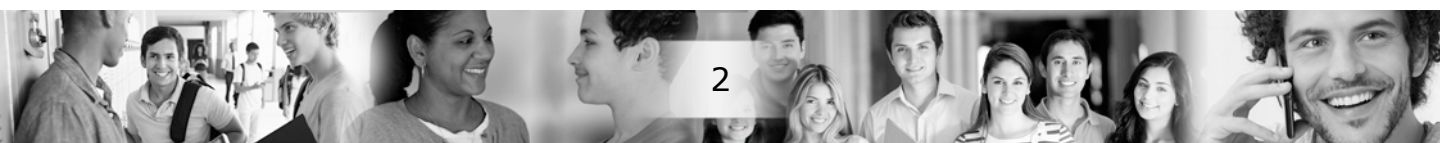
Kim: No, both stand for the same imperative paradigm; the difference is that the procedural programming uses the concept of blocks to set the program, I mean, blocks are groups of statements that contain a whole procedure, while in the procedural programming the statements are separated.

Mark: I see. To conclude, Kim. I wish you could give us some examples of imperative languages.

Kim: Sure, Java and .Net Framework are clear examples of imperative languages, as both use variables that can be either internal or external.

Mark: Does that mean that all imperative languages use variables?

Kim: That's correct, all use them, especially high-level imperative languages such as Java or C#.





Let's Practice!

All of them are imperative languages!

Instructions: Read the paragraph, and select all the correct answers.

1. Which of these sentences are true about imperative languages?
 - a. Most programs need to share similar paradigms, as all use imperative languages.
 - b. We expect to use both, Java and .Net Framework, as they are imperative languages.
 - c. We agree to believe that programming languages are all the same because all belong to the imperative paradigm.
 - d. I decided to use Java, C#, and .Net Framework because all are imperative languages.
 - e. You need to understand that both, JavaScript and HTML are imperative languages.



Let's Practice!

Let's review what imperative languages are.

Instructions: Select the correct answer to complete the sentences.

1. I like using C# and Java because _____ are imperative programming languages.
(Hint: C# and Java are only two elements.)
 - a. some
 - b. many
 - c. both
 - d. all
2. Most imperative languages work similarly, as _____ use logic commands in their programming.
(Hint: We are talking about a whole group of languages.)
 - a. every
 - b. each
 - c. all
 - d. none
3. In terms of programming paradigms, _____ are valid and useful.
(Hint: Every single paradigm.)
 - a. some of
 - b. all of
 - c. both
 - d. all
4. Variables are one of many important factors to determine an imperative language, but not _____ make use of them to achieve its goals.
(Hint: A whole.)
 - a. all
 - b. both
 - c. none
 - d. some



Extension Activity

Let's see how much you know about imperative languages.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about two imperative programming languages.
- Write a brief description of each of these languages.
- Use indefinite pronouns, verbs with infinitives, and simple present.
- Share your work with your partners.

Self-Assessment

- I look for information about programming languages.
- I write a brief description of those languages.
- I use indefinite pronouns, infinitives and present simple.
- I share my work with my partner.

Yes

No

☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- command** (noun): an order or instruction
 e.g. An imperative programming language is, basically, a list of commands for the computer to perform.
- paradigm** (noun): a model, or parameter to do something
 e.g. For today's program, we decided to bring into discussion the imperative languages paradigm in computer science.
- logic** (noun): a credible way of thinking reasonably
 e.g. That's why programmers fix logic statements orderly and descriptively for the computer to execute.
- enable** (verb): to make something possible by itself
 e.g. The imperative paradigm is very important because it enables all elemental computer functionalities to work.
- goal** (noun): the expected results of an action.
 e.g. The imperative paradigm focuses on the processes the program performs rather than on the goals that the program needs to achieve.



Learning Points

Indefinite Pronouns

Indefinite pronouns are words that replace unspecific general nouns; use them when you want to refer to an overall noun.

There are different **indefinite pronouns** for either singular or plural nouns. Today we are going to focus on two common indefinite pronouns: **All** and **both**.

All

You use **all** when you are pointing to the whole amount of the noun you are talking about.

Example:

*We can't change one single program without modifying the others, as **all** need a special code. (All the programs.)*

***All** you need to understand how Java works is in the manual I gave you. (All means all the tools)*

Both

Use both when you are relating two similar ideas; both is a plural pronoun, as it refers to two things.

Example:

- **A:** *I don't know what to use to program this app, Java or C#.*
- **B:** *I think **both** are good options. (both means Java and C#)*
- *A program is made of **both** the design and the codification.*

Examples

- *No, both stand for the same imperative paradigm; the difference is that the procedural programming uses the concept of blocks*
- *That's correct, all use them, specially high-level imperative languages such as Java or C#.*



Learning Points

Infinitives after Certain Verbs

There are verbs that introduce other actions; Those verbs can introduce the actions with the infinitive, with the gerund, or, in some cases, with both of them. Let's have a look at the **verbs** that go with **infinitives**.

There are many verbs that use the **infinitive** to introduce another action; some of those verbs are: want, need, agree, decide, and except, among others. Remember that you always need to use **the infinitive** after those verbs.

Note that the word **infinitive** refers to **the base form** of the verb accompanied by the preposition **to** as in **to buy** and **to program**.

The structure is very simple:

Subject + Verb + Infinitive + Complement

For example:

*I **want to know** the difference between imperative and declarative languages.*

*She **needs to implement** an imperative language to make the program works.*

*We **expect to finish** programming this app soon.*

Examples

- *Kim, a software programmer, agreed to join us and help us to understand this topic.*
- *We decided to bring the imperative languages paradigm in computer science into discussion.*

